

Article

Artificial Intelligence Chatbots as Assistants for Media Users: The Cases of El País and El Espectador

Gema Sánchez-Muñoz ¹, Isabel García Casado ²  and David Varona Aramburu ^{1,*} 

¹ Department of Journalism and New Media, Faculty of Information Sciences, Universidad Complutense de Madrid, 28040 Madrid, Spain; gsanchez01@ucm.es

² Department of Online, Economics, Business, International Relations and Communication, Faculty of Economics, Business and Communication, Universidad Europea de Madrid, Villaviciosa de Odón, 28670 Madrid, Spain; isabel.garcia@universidadeuropea.es

* Correspondence: davarona@ucm.es

Abstract

In recent months, some media outlets have been launching artificial intelligence-based chatbots that serve as assistants to users in their search, selection and consumption of content. This research analyses two such examples: Vera, a conversational assistant launched by the Spanish newspaper El País, and the model used by the Colombian newspaper El Espectador, which operates on the WhatsApp platform. Both chatbots share the same approach: they are tools designed for users to interact with newspaper content. This interaction takes place through natural language conversations: the technology understands ‘users’ questions or requests and provides answers based on the content hosted in the newspapers. This changes the way media content is explored. We are moving from a paradigm centred on search engines and keywords to one in which conversation determines the discovery of content. The research analyses the results of these two pioneering experiences in the Spanish-language media. The aim is to understand the extent to which they are changing the relationship with content and how they are affecting the media.

Keywords: artificial intelligence (AI); conversational chatbots; WhatsApp; natural language; digital media

1. Introduction

The expansion of Large Language Model (LLM) technology is having a deep impact on journalism and the media. Surely, the most visible part of this impact is the proliferation of so-called chatbots or conversational assistants, some already as popular as ChatGPT, Gemini, Claude, or Perplexity, to name just a few. These assistants have multiple functions, and the public has also begun to use them to find content that is then recast into new texts by the chatbot.

It is precisely this use, which we could define as an “information search engine-synthesiser,” that is of interest in this research. We aim to find out how some media outlets are integrating artificial intelligence chatbots into their websites, thus offering a new way of accessing content, as many companies in other sectors, especially in the field of e-commerce, have done. With these tools, instead of navigating through the menus and sections of digital newspapers, or instead of searching using conventional search engines, users hold conversations with artificial intelligence assistants through prompts. Based on the instructions given by users in these conversations, the assistants search for information in the newspaper’s database, rework it and offer it in a synthesised form.



Academic Editors: Eglée Ortega
Fernández and Carmen Llovet

Received: 30 December 2025

Revised: 14 February 2026

Accepted: 10 March 2026

Published: 18 March 2026

Copyright: © 2026 by the authors.
Licensee MDPI, Basel, Switzerland.
This article is an open access article
distributed under the terms and
conditions of the [Creative Commons
Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

To analyse this emerging and innovative way of interacting with digital media content, we have selected two recent examples: the artificial intelligence chatbots implemented in the newspapers El País (Spain) and El Espectador (Colombia).

These chatbots are two of many examples that have appeared in recent months. El País launched its product, called Vera, in March 2025 (El País, 2025), while El Espectador presented its product just one year earlier (Vega, 2024). They are part of a significant contingent of similar initiatives, including the launch by Time magazine, which rolled out its artificial intelligence agent in November 2025 (Howard, 2025). Other major media outlets that have ventured to test this technology or are implementing it include Agencia EFE, Financial Times, Süddeutsche Zeitung and Bild.

In this research, we focus on two of these media outlets that are committed to artificial intelligence assistants. Both are leaders in the international press in Spanish and stand out for their paradigmatic and referential role for their national industries and those of the countries over which they exert influence.

2. Impact of AI on Journalism

The abundant bibliography published in recent years on journalism and artificial intelligence (Ioscote et al., 2024; Manisha & Acharya, 2023; Xie, 2021; Parratt-Fernández et al., 2021; Tüñez-López et al., 2021) demonstrates the interest (and concern) that this field arouses both in academic circles and in the journalism profession itself.

There is no specific point at which these studies converge. The incipient and widespread emergence of this technology in the media (Tejedor & Vila, 2021; Chan-Olmsted, 2019), as well as the uncertainty surrounding the possible development of artificial intelligence tools and their use in the media, make it difficult to identify possible proposals that would establish AI as a useful tool. And we must not forget the negative consequences of this technology, such as the feared replacement of journalists (Rentoul, 2020; Moran & Jawaid, 2022; Londoño-Proañó & Buele, 2025).

Within this framework of uncertainty, the implementation of AI in the journalism industry shows a strong progress. For example, in the development and application of machine learning. On the one hand, it is used for journalistic production; on the other, for studying audience behaviour using “machine learning algorithms” (De-Lima-Santos & Ceron, 2022). Something similar is happening with link positioning in search engines. In principle, media SEO has benefited from generative artificial intelligence tools in terms of positioning and readability of content that favours traffic to the media (Apablaza-Campos et al., 2025). However, with the arrival of AI MODE, launched by Google in October 2025 (Google, 2025), it remains to be seen to what extent search and positioning, and consequently traffic, will be affected.

On the other hand, given the ethical importance of the profession, emphasis is being placed on the issue of journalistic ethics in the use of AI (Oh & Jung, 2025). The debate is on the table, but it is necessary to explore deeper into the proper or improper use, as well as the positive and negative aspects of such use (Cools & Diakopoulos, 2024). However, concern about ethical and moral issues has grown exponentially in recent years as the use of AI has been implemented in many contexts (Zanzotto, 2019), and especially in newsrooms (Chuan et al., 2019). The question then arises as to whether we are looking at a simplified view of the repercussions of the use of AI in journalism, without considering the ethical and bias issues that may exist in the information generated by these tools (Lewis et al., 2019). For journalists, AI has arrived, above all, to save them from tedious and repetitive work, so that they can concentrate on the information, on the purely qualitative (Noain-Sánchez, 2022).

However, the work of the editor is undoubtedly the most difficult obstacle facing the new model of journalistic production. Practice has shown that the narrative produced by

AI platforms, while not an exact replica of human writing, is similar enough to be a viable alternative (Lewis et al., 2019). However, most authors propose AI as an auxiliary tool, always with human supervision (Noain-Sánchez, 2022).

To all this, we must add the skill that information professionals achieve with the tools they work with. A study by Terol (2023) highlights some shortcomings, starting with the gap between the speed at which technology advances and the industry's ability to adapt to that pace. This is one reason why communicators are seen as ill-prepared, with an urgent need for renewal. But, at the same time, there is also a lot of resistance to the implementation of artificial intelligence tools in newsrooms. Along the same lines, Romera et al. (2024) describe the optimal and efficient use and learning of these tools as "a major challenge for digital newsrooms".

2.1. Artificial Intelligence in Communication in Spain and Colombia

The United States and Europe lead the way in the use of AI applications in the journalism industry, as the high cost of this technology limits its use in newsrooms in regions such as Latin America (De-Lima-Santos & Ceron, 2022). Despite these difficulties, more and more media outlets are incorporating these practices into their strategies to optimise, simplify and make their production processes more effective (Díaz Noci, 2023; Pérez-Seijo et al., 2020; Trapova & Mezei, 2022). There is no doubt that we are in a period of metamorphosis towards automation in news generation (Trejos-Gil & Gómez-Monsalve, 2024), something that is very palpable in Spanish newsrooms and is beginning to be so in countries such as Colombia. However, in the case of the South American country, there is hardly any research focusing on the use of artificial intelligence in the media.

The few studies related to the subject in Colombia can be found in some works that focus on the importance of AI in political communication. For example, a precedent for studies on automation processes is the case of Twitter and the 2014 Colombian elections. The objective of the study by Cerón-Guzmán and León-Guzmán (2016) was to identify the trend in public opinion and the expansion of subjectivity among citizens marked by misinformation spread by social bots.

The proliferation of bots, used for no other purpose than to spread misinformation, is one of the reasons why the journalism profession has been forced to take on the task of verification as a new service to the public. In fact, it is with AI itself, through machine learning processes, that specialist media outlets such as Colombiacheck and La Silla Vacía detect and combat disinformation (Rodríguez-Pérez, 2021).

In Spain, we find more studies addressing the use of AI in newsrooms. One of the most recent is the work carried out by Sarrionandia et al. (2025) on the analysis of AI applications in the media. In it, they find that 56.5% of the professionals surveyed from different media outlets use some type of artificial intelligence tool. The purposes of this use range from qualitative (obtaining data and results) to proofreading and revising texts. However, it almost always appears as a support tool. Among the findings of this research, it is also worth noting that the use of AI applications to generate journalistic texts is not common, although it is more frequent in the generation of multimedia content.

For their part, Romera et al. (2024), in their study on the online media outlet El Español, conclude something similar to what has already been stated. AI is seen as an aid, a "facilitator," but in no case a replacement for journalists. The challenge lies in finding the balance "between automation and human intervention".

With these caveats in mind, the truth is that the use of AI in journalism is becoming widespread. Without attempting to be exhaustive, its use in Spain alone is already widely documented in diverse practices such as the verification of disinformation (Fernández Barrero & López Redondo, 2022), social media management (Romera et al., 2024) and data

visualisation (Garriga et al., 2024; Pérez-Montoro & Lopezosa, 2026). Taking a more general view, we see that its use is recorded in content distribution, automated information and documentation collection, and automated information production (Mayoral-Sánchez et al., 2023; Mayoral-Sánchez et al., 2024).

The work of Mayoral-Sánchez et al. (2023) points to this use of information automation and adds another: audience engagement, as Perreault and Ohme (2025) suggested. These are precisely the two pillars on which the object of this research rests: the use of artificial intelligence chatbots in digital media.

2.2. Chatbots: An Evolution

Chatbots burst onto the journalism scene a decade ago, when many media outlets began experimenting with the first versions of these tools, which did not yet feature artificial intelligence. According to Ford and Hutchinson (2019), those chatbots were computer programs that responded to user information by simulating a human in conversation. In other words, they were tools that fell within what authors such as Trillo-Domínguez and Alberich-Pascual (2017) defined as conversational journalism.

The use of chatbots by the media was very varied and served many purposes. One of the best-known classifications of chatbots according to their use is that of McKelvey and Dubois (2017). These authors distinguish, for example, between amplifier chatbots, transparency chatbots, and servant bots or butlers. Lin et al. (2023) also found that the first chatbots were used for five purposes: including technical improvement, context maintenance, business support, educational objectives, and other specific objectives.

These first-generation chatbots were based on simple programming. Interaction with users took place through predefined actions which, as they occurred, delivered certain contents. In this sense, they were relatively simple programmes that could run on social networks (such as Facebook Messenger) or messaging platforms, such as WhatsApp or Telegram. They have also been widely used on websites, particularly in e-commerce environments (Gao et al., 2025), but to a lesser extent in journalism.

Among the advantages of these early tools, Herrero-Diz and Varona-Aramburu (2018) pointed out the importance of proximity to the user and the possibility of personalising the message, which generated a feeling of attachment among the public. This idea is consistent with the contributions of Sánchez Gonzales and Sánchez González (2017), who have discussed the “emotional connectivity” that occurs between chatbots and users. It also coincides with the work of Ford and Hutchinson (2019): “Users expressed appreciation for the non-intimidating, non-threatening tone of the bot”.

However, the evolution of those pioneering developments was not easy, which affected the implementation of chatbots in all media. For example, their evolution was affected by the high costs involved in their deployment: on the one hand, the tool itself and, on the other, the need for new specialised human staff to supervise it. Similarly, there was a lack of consistency in the text produced by the chatbot compared to that produced by the journalist (Belair-Gagnon et al., 2020). Ford and Hutchinson (2019) pointed out perhaps the weakest point when they found that one of the major differences between those early chatbots and today chatbots was that “testing the functionality of free text questions to the bot, we realised that its ability to respond to questions is limited.”

Thus, it soon became clear that chatbots would make more sense if artificial intelligence were integrated into them. Herrero-Diz and Varona-Aramburu (2018) indicate that, with the incorporation of AI into chatbots, feedback is also obtained, since, as a result of this improved interaction, it is possible to gain a deeper understanding of user behaviour and “some clues about their identity”.

2.3. Generative Artificial Intelligence Chatbots

That first stage of chatbots, which gave rise to the first conversational journalism, has given way to a new situation in which artificial intelligence proposes a new type of interaction between the public and the media. In an evolution of increasingly fluid and realistic conversational communication between machines and people, AI increases these dialogue abilities to levels of human familiarity. From this perspective, [Miguel and Cabrera \(2020\)](#) defined these chatbots as tools that, with the use of AI, serve as interlocutors with people.

Obviously, this exchange of messages between cybermedia and people is not limited to the arrival of chatbots. Since the first decade of the 21st century, the digital press had established relationships with subscribers and users through tools such as forums and blogs, which not only provided information to users but also facilitated an important exchange of opinions between interlocutors ([Pano, 2012](#)). In this context, for [Nordberg and Guribye \(2023\)](#), the figure of the journalist is core in the equation of conversational journalism to generate dialogue with the audience. Editors and readers interact, the authors argue, in the form of natural and open conversation.

In this evolution, chatbots equipped with artificial intelligence must achieve the conversational ability of humans thanks to technologies such as NLP (natural language processing) ([Lin et al., 2023](#)). However, although digital media outlets are introducing chatbots into their environments, it is important to note that users still do not use them very intensively, as only three per cent of Spanish users say they have used them to obtain information ([Vara-Miguel, 2025](#)). [Ufarte-Ruiz et al. \(2023\)](#) have documented the onset of this transformation and have already referred to it as “the first newsrooms without journalists”.

3. Research Objectives

In the context described above, this research aims to explore the emerging field of artificial intelligence assistants within digital media. To this end, we will start from the following:

Research question: How are some Spanish-language digital media outlets and their users already using artificial intelligence chatbots?

To find answers to this initial question, the research sets out the following objectives:

1. Objective 1—To observe and understand the functioning of the artificial intelligence chatbots of El País and El Espectador from the user’s point of view.
2. Objective 2—To understand the interest of these two media outlets in implementing these tools.
3. Objective 3—To learn about and evaluate the initial results of the implementation of these assistants, as well as their potential in the media industry.

4. Materials and Methods

This study is situated within a conceptual framework related to the figurational approach ([Elias, 1995](#)) applied to the automation of communication. Accordingly, newsrooms are understood as “hybrid figurations” in which humans and AI systems are interdependent. This way of approaching the issue of AI in journalistic newsrooms has been examined from a figurational perspective in several studies, such as those by [Hepp et al. \(2023\)](#). These authors argue that it is an approach “suitable for researching communicative AI” because it does not construct contradictions between the individual and society and because it focuses particularly on issues such as change and transformation and on the relationships between technology and, in this case, professionals and audiences.

Another study that analyses the implementation of technologies from a figurational perspective is that of [Dahlman et al. \(2021\)](#), who conceptualise algorithms as organisational

figurations, that is, sociotechnical arrangements configured in spatio-temporal processes. This approach has also been applied in studies of technology implementation such as those by [Adams \(2024\)](#), [Ali et al. \(2024\)](#) and [Webster \(2025\)](#).

On the basis of this approach, a case study is proposed which, in turn, comprises two research instruments. First, interviews were conducted with those responsible for developing and implementing the chatbots of El País and El Espectador. Second, the functioning of these chatbots was observed and analysed in order to gain a better understanding of their possibilities and limitations.

Both instruments fit well with the sociotechnical or figurational approach, as they enable us to capture both practices and meanings within human–technology figurations, as occurs in interactions with media through chatbots. On the one hand, observation captures “what is done” in sociotechnical figurations (routines, actual uses of technology, implicit hierarchies), as proposed by authors such as [Denny and Weckesser \(2022\)](#) or [Chand \(2025\)](#). On the other hand, semi-structured interviews make it possible to explore in depth the meanings, justifications and strategies that actors attribute to these practices, in line with the contributions of [Kallio et al. \(2016\)](#) and [Knott et al. \(2022\)](#). According to [Alonso \(1999\)](#), “the research interview seeks, through the gathering of a set of private knowledges, to construct the social meaning of individual conduct or of the reference group of that individual”. Within this framework, the semi-structured interview is a highly suitable technique and, therefore, widely used in research on journalistic issues. It makes it possible to maintain an analytical and investigative structure while remaining flexible enough to adapt the design to the needs of each moment and circumstance.

This adaptability is particularly valuable in journalism research, where the aim is often to explore the multifaceted realities of professional practice, the motivations and challenges faced by journalists, and the broader sociocultural context ([Junnier, 2024](#)) in which journalism operates. In addition, it is a highly appropriate technique for exploring issues related to professional practice ([Husband, 2020](#)) that are often emergent, as in the case addressed here, which concerns a highly topical and perhaps short-lived phenomenon. Moreover, semi-structured interviews are well suited to mixed-methods designs, allowing for the triangulation of qualitative perspectives with quantitative data and supporting a comprehensive understanding of journalistic phenomena ([McIntosh & Morse, 2015](#)).

4.1. Interview Protocol

The interviews were conducted in May 2025, always in the form of video calls via Google Meet, a tool that enabled the conversations to be recorded. They were carried out by two members of the research team: one interviewed the representatives of El País and the other those of El Espectador.

The key individuals involved in the implementation of the chatbots examined in this study took part in the interviews (N4) (see Table 1). The duration of each interview ranged from 30 to 60 minutes.

Table 1. Key actors interviewed in each outlet. Own elaboration.

Key Actors Interviewed at El País	Key Actors Interviewed at El Espectador
Head of Product Development	Director of New Technologies
Head of Social Communities	Multimedia Editor

A guide was first prepared in order to carry out the interviews (see Appendix A). This interview guide was structured into three analytical sections. The guide was constructed on the basis of the analytical categories that underpin the study and that were defined to address the previously formulated objectives. These categories, six in total (see Table 2),

emerge from two sources: the review of scientific literature related to the implementation of new technologies in the media, and the detailed observation of the chatbots that constitute the objects of study in this research.

Table 2. Analytical categories used, their relevance to the study and evidentiary definitions. Own elaboration.

Analytical Category	Relevance to the Study	Evidences
1. Outlet's motivation for implementing the chatbot	To understand the strategic, editorial or technological reasons that drove the development of the chatbots.	Statements will be considered evidences in which the interviewee explicitly sets out the reasons justifying the adoption of the chatbot, including references to the outlet's needs (innovation, efficiency, automation), competitive pressure, sector trends, audience expectations or alignment with the digital strategy.
2. Objectives of the chatbot	To analyse the function that each chatbot seeks to fulfil (to inform, build loyalty, personalise, distribute, etc.).	Fragments in which the purpose of the chatbot is explicitly described will be coded as evidences, such as improving the user experience, increasing engagement, personalising content, automating responses, capturing traffic or supporting journalistic work.
3. Technology used	To ascertain which language models have been employed, their technical configurations and possibilities for integration with other platforms.	Mentions of tools, platforms or technological infrastructures (language models, APIs, specific software), as well as descriptions of technical functionalities, integrations (CMS, social networks, apps) or technological limitations of the system, will be identified as evidences.
4. Development and implementation process	To determine the project phases, involvement of internal and external teams, duration.	Discourses describing the chatbot creation process will be considered evidences, including phases (design, development, testing, launch), actors involved (internal teams, external providers), execution times and difficulties encountered during implementation.
5. Reception by users	To understand the opinions, level of use, consultation habits, suggestions or complaints generated by the use of the chatbots.	References to user perceptions, behaviours and feedback will be coded as evidences, including levels of use, interaction patterns, positive or negative evaluations, complaints, suggestions or satisfaction metrics.
6. Evaluation of the impact on the outlet	To determine a positive impact in terms such as web traffic, loyalty, monetisation, competitive differentiation. To determine a negative impact in terms of development and usage costs, perceived errors, possible conflicts with the audience or newsroom.	Statements evaluating the consequences of chatbot use will be considered evidences, both positive (increased traffic, engagement, revenue, competitive positioning) and negative (costs, errors, misinformation, internal tensions or user rejection), including metrics or qualitative perceptions.

The interview guide was developed through an iterative process conducted by the research team. Each member of the team proposed a set of topics and questions, which were then discussed in order to distil the themes that would make up the final interview guide. It is considered that the topics included in the guide are sufficient to achieve the level of saturation sought in an exploratory study.

Once the guide had been drawn up, the next step was the selection of the sample. In this case, the selection of key actors (Table 1) took place following prior conversations with the content management of both media organisations. These initial contacts led us to these key individuals who, although constituting a small sample, are responsible for the configuration, development and implementation of the chatbots in the two companies.

4.2. Content Analysis

Subsequently, the interviews were transcribed and subjected to a content analysis process. This technique makes it possible to examine messages and representations in the media in a systematic and objective manner, identifying recurring patterns, themes and approaches. It also facilitates the inference of knowledge about the conditions under which information is produced and received, as well as the assessment of trends, biases or news values in the discourse of journalism and media professionals (Wheatley, 2021).

To carry out the content analysis, work was conducted on the basis of the set of analytical categories described above, which operated as a codebook. These six analytical categories make it possible to capture both the strategic logic of the media and the editorial, technological and organisational implications of these developments. Their selection responds to the need to address the phenomenon from a comprehensive perspective, consistent with the literature on innovation in online news media, news personalisation and the adoption of algorithmic technologies in digital newsrooms (Noain-Sánchez, 2022; Niño & Borges, 2025) (see Table 2).

The categories were operationalised through evidentiary definitions that enable the identification of units of meaning in the verbatim material, thereby ensuring consistency in the coding process. In addition, inclusion and exclusion criteria were defined for each category in order to delimit the units of analysis. Fragments in which interviewees made explicit reference to the phenomenon under examination were included, while tangential, ambiguous or not directly related mentions to the dimension under study were excluded. Furthermore, in cases where the same fragment could be assigned to more than one category, a criterion of semantic predominance was applied, assigning it to the category that best represented its main meaning. This procedure contributes to strengthening the reliability and replicability of the analysis.

Firstly, the media outlet's motivation for implementing the chatbot allows us to contextualise the project within the overall strategy of the journalistic organisation. The introduction of conversational assistants is not a technologically neutral act, but rather tends to respond to structural pressures within the media ecosystem—such as audience fragmentation, information overload or dependence on external platforms—as well as editorial or innovative positioning objectives. Based on this category, we can identify whether the chatbot is conceived as a defensive response, an experimental venture, or a central element of the media outlet's value proposition.

Secondly, analysing the chatbot's objectives is key to understanding the role assigned to it within the media outlet's information system, as, beyond their technical appearance, these assistants can be geared towards various purposes: facilitating access to published information, strengthening loyalty, personalising the user experience, or improving content distribution, among others. This category therefore allows us to assess the degree of alignment between the design and perceived performance of the chatbot and the editorial and business goals pursued by the organisation in creating it.

The technology used constitutes the third analytical category, which is essential for understanding the capabilities and limitations of the chatbot. The choice of certain language models, technical architectures, or integration systems with databases and content managers conditions both the quality of the responses offered and editorial control over the

system. From an academic perspective, this category allows us to link empirical and observational analysis with debates on automation, technological dependence and algorithmic transparency in digital journalism.

Fourthly, the development and implementation process is incorporated as a category to address the organisational dimension of the project. The literature on media innovation and its collaborative application (García-Avilés, 2021; Xiao et al., 2025) emphasises that the success or failure of new tools depends largely on how the teams involved are organised, on collaboration between journalists and technicians, and on the newsroom's ability to integrate the chatbot into its routines. In this way, we identify models of technological governance and internal dynamics that are decisive for the sustainability of a technology that, after its initial implementation by the media outlet, will inevitably evolve and push editorial teams to embrace this continuous transformation.

User reception constitutes the fifth category, focusing on the actual interaction between the chatbot and the audience. Given that these assistants are presented as reader-oriented tools, it is essential to examine their degree of adoption and usage habits, as well as the perceptions, suggestions or resistance expressed by users. This category connects technological analysis with initial audience data to assess the extent to which the chatbot responds to real information needs and whether it generates friction—and if so, what kind—in the consumer experience.

Finally, the assessment of the impact on the media is presented as a synthetic category that integrates both positive and negative effects. From an organisational and editorial point of view, it is important to assess whether the chatbot contributes to increasing traffic, reinforces loyalty, opens up avenues for monetisation or contributes to the media's differentiation in a highly competitive environment such as the media. Associated costs, identified errors, possible conflicts with the audience, and internal tensions in the newsroom are also considered in order to achieve a balanced and critical view of the real scope of this innovation.

Given the small number of interviews, manual coding was chosen. For this purpose, a simple codebook was used in which each category corresponds to a code. Two members of the research team coded the interviews independently, and the third member acted as an arbitrator to resolve discrepancies, following a process similar to that described by Mayring (2014) and employed in studies such as that of Dengel et al. (2023), which specifically worked with chatbot support. Several examples of coding applied to interview verbatim material can be seen in Table 3.

Table 3. Examples of coding applied to interview verbatim material. Own elaboration.

Examples of Responses/Verbatims	Coding Categories
"We have other functionalities that are tied to artificial intelligence, and what we wanted was to leverage the tool to improve the way people can access our content."	- Objectives of the chatbot
"The language and everything we did to make navigation easy was precisely designed so that a 60-year-old person, who perhaps does not handle a lot of technology, could navigate it easily and understand what the system does."	- Reception by users - Development and implementation process
"We have found interesting things; for example, people do not always ask in the same way, or there are questions that are very particular to each user."	- Reception by users
"We see that voice notes are proving useful, especially for quick queries."	- Evaluation of the impact on the outlet

4.3. Observation

Observation was also employed to assess the value of the proposal. However, since chatbot responses vary depending on the user, the prompts posed, technical knowledge, conversation history and numerous other factors, observation becomes more complex and demanding.

To conduct this observation process, a protocol was established and implemented over four days. During these four days, both chatbots were subjected to the same battery of questions (Table 4), repeated identically in the morning and afternoon sessions.

Table 4. Questions posed to the chatbots during observation. Own elaboration.

Questions for Chatbot Observation:
Hello. What will the weather be like today in Madrid?
Hello. What are the most-read news stories of the day in El País?
Hello. Tell me the most up-to-date information on Donald Trump.
Hello. Tell me the latest La Liga football results.
What is the latest scientific discovery published by El País?

The aim of this protocol was to understand how the chatbots respond to user questions depending on the time of day. The questions were repeated to ensure that the observation conditions were as similar as possible.

Thus, one team member observed the Vera chatbot from El País, while another did the same for the El Espectador chatbot. In both cases, observation was carried out using Chrome browsers, with geolocation enabled, after logging into each outlet and subscribing to their Premium paid memberships.

During the four days of observation, from Sunday to Wednesday in a week of February 2026, the questions were asked in both morning and afternoon sessions. Subsequently, the responses from each tool were stored in the form of screenshots over the four days of observation, which made it possible to compare identical questions and times.

Once all this material had been collected, the three team members analysed it according to a rubric that had been previously prepared (see Table 5). This rubric was generated partly from the categories used to analyse the interviews and also from what the media outlets themselves have stated about these services. That is, what they have announced as expected from them in their interaction with the public.

Table 5. Example of the rubric used to observe the chatbots and examples of the results from an observation period. Own elaboration.

Question 4—Day 3, Afternoon Session	El País (Vera)	El Espectador
Do they respond to the question?	Yes	No
Is the response sufficient?	Yes	No
Is navigation required to obtain the response?	No	Yes
Do they include links in the response?	Yes.	Yes, but unrelated to the question
Is adaptation of the response to the user's location evident?	No.	No
Is there variation in responses across days?	Yes, it has changed.	No
Do they personalise based on user knowledge?	Not observed.	Not observed.

5. Results

The case studies, carried out using the techniques described above, have yielded interesting results that provide a better understanding of these media outlets' interest in implementing artificial intelligence chatbots and how they work. Below, we will analyse the results for each of the two cases, first placing them in the context of their origin.

The implementation of Vera, the commercial name of the artificial intelligence chatbot deployed by El País in March 2025, arose from an agreement between the Spanish newspaper and the company OpenAI, owner of the well-known ChatGPT assistant (El País, 2024). This agreement stipulated collaboration between the two companies. On the one hand, OpenAI could use the newspaper's contents to train its deep language models; on the other, El País could have privileged access to OpenAI's technology to launch initiatives and products such as the Vera assistant.

As mentioned above, the chatbot was launched on 25 March 2025. In the first phase, it was tested with a small group of users, and surveys were then conducted to evaluate its performance and make adjustments. It was then made available to subscribers.

Operationally, it appears as a blue button that “floats” on the El País website or app and can only be used by premium subscribers to the newspaper. When this button is activated, a box appears in which Vera is displayed as a typical chatbot, with an invitation to participate: “Where shall we start? Ask me anything” (see Figure 1). From there, users can ask questions and chat with Vera, whose answers will always be based on the contents of El País.



Figure 1. Vera's home page on El País.

From there, the conversation allows the user to request information in successive questions or prompts. The chatbot's responses are always in Spanish, although it accepts questions in English and other languages (see Figure 2). Formally, they combine texts reworked by artificial intelligence with links to contents that can expand on the information. In addition, various topics are suggested for users to explore if they do not want to make a query.

From El País, both in the interviews and in the communications issued to launch the chatbot (El País, 2025), emphasis is placed on the idea that Vera uses the content published by the newspaper on its website in its responses. Our observation has confirmed that this is true, but it is not always fulfilled with equal effectiveness. In many cases, a seemingly straightforward question such as “What are the most-read news stories of the day in El País?” does not receive an elaborated response, but simply refers to a consultation link (see Figure 3). Similarly, a straightforward question requiring La Liga football results is answered only on some occasions, whereas at other moments during the observation period the chatbot either fails to provide an elaborated response or makes an error and, instead of offering results, provides a schedule of future events (see Figure 4). In general, we encountered this problem with all the questions: at certain specific moments, the chatbot was unable to respond, whereas at others it did provide an AI-generated elaborated response.

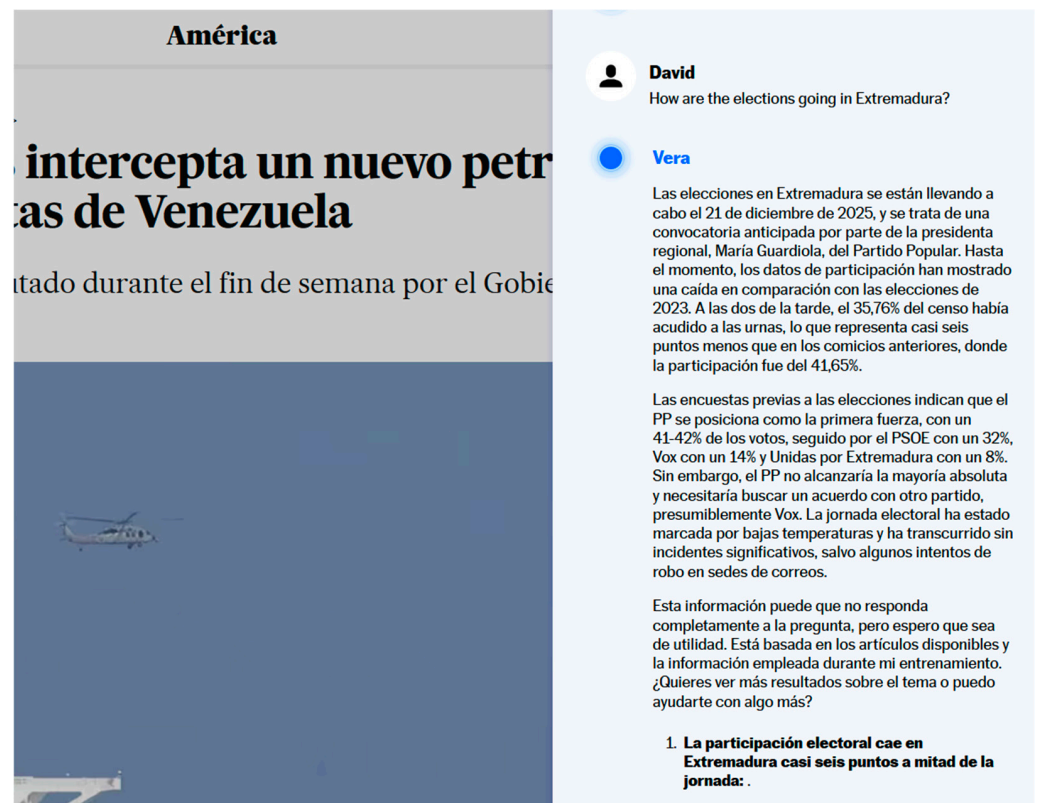


Figure 2. Example of interaction with El País chatbot.

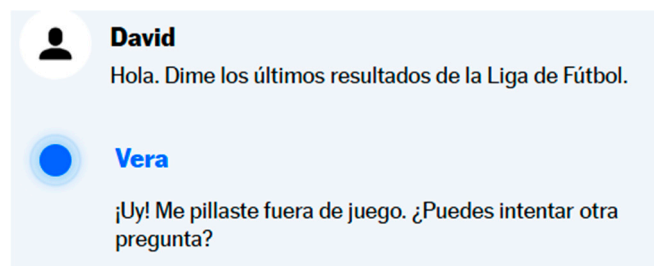


Figure 3. Examples of failed responses.

In the case of the Colombian newspaper El Espectador, its AI-powered news chatbot has been implemented as a progressive initiative aimed at adding value to digital subscriptions, as this tool is a differentiating factor from the competition that can guarantee subscribers' acquisition and loyalty.

The strategy has been developed in two phases: in the first, launched in March 2024, the chatbot was integrated into the web edition of the media outlet (see Figure 5) and from the outset was accessible exclusively to subscribers. Users can interact with it in natural language to consult recent news and featured content (traffic, weather forecast, etc.); When activated, readers can click on suggested queries ("I need the latest news," "I'm interested in the most read articles," "I want to know about...") or ask open-ended questions about current events, which the chatbot answers with responses generated from the newspaper's own archive and journalistic output, accompanied by contextual links. Access is limited to a maximum of twenty queries per user per day, which reinforces its type as a premium service and opens up the possibility of charging heavy users who quickly exhaust this query limit a higher subscription fee in the future.

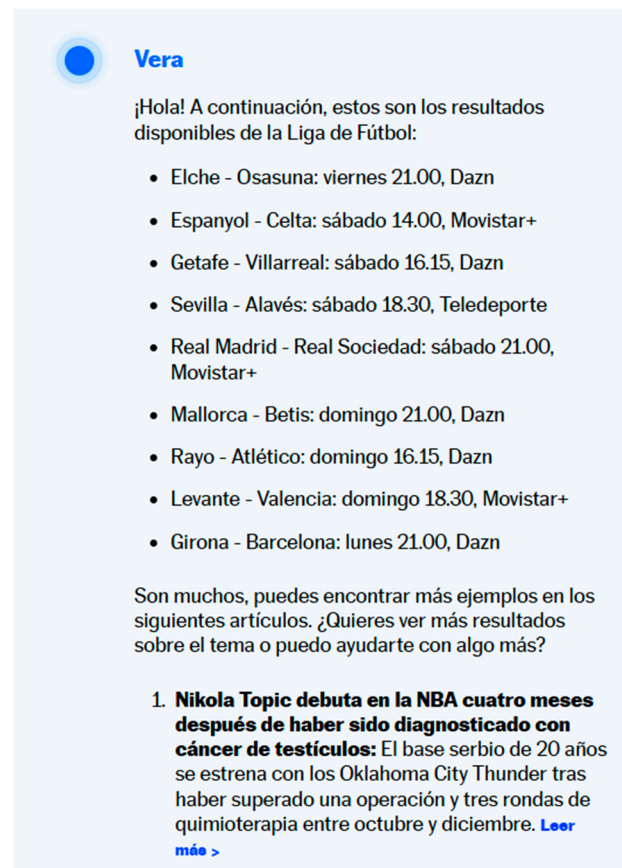


Figure 4. Examples of failed responses.



Figure 5. In the right side, chatbot for subscribers on El Espectador website.

In the second phase, launched in March 2025, El Espectador expanded the scope of the assistant by deploying it on the WhatsApp messaging network, thus transferring the conversational experience to an everyday environment widely used by citizens in Colombia. This channel maintains the logic of exclusivity for subscribers and functions as an information assistant that facilitates news consumption without the need to directly access the website.

The WhatsApp chatbot functions as a contact that people save on their phones, although it initially presents them with a registration form asking for some details, such as email address and gender. As shown in Figures 6 and 7, once it is verified that the user is a subscriber, the tool begins to provide information, with responses in natural language in both text and voice notes. They have also created a function for the chatbot to send direct messages, allowing them to issue alerts about important news or events.

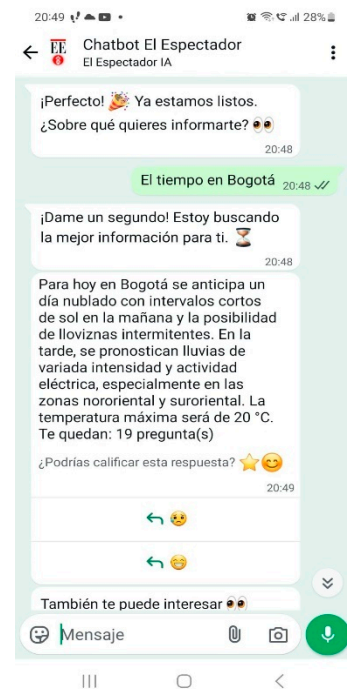


Figure 6. Conversation between a user and El Espectador’s chatbot.

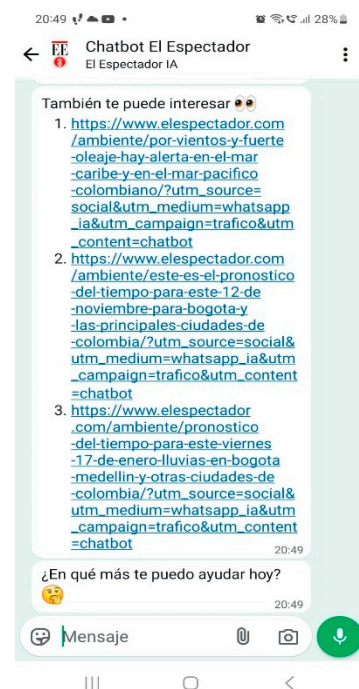


Figure 7. Example of proposal for complementary content by the assistant.

The newspaper team sees this as an inbound strategy, with all its positive aspects: it is the reader who actively seeks out the news to satisfy their information needs, making the use of invasive delivery techniques unnecessary.

The stated objective of the creators and managers of El Espectador’s chatbot, as expressed in the interviews regarding its interaction with readers, is that this interaction should take place in natural language; however, the conversational assistant fails to understand some straightforward questions (such as “What are the most-read news stories of the day in El Espectador?”) and is unable to provide any response, remaining stuck in latency for several seconds before displaying “I have no information about the most-read news

stories of the day in El Espectador” (see Figure 8). Although one of the pre-established queries it offers when first activated is “Check most-read content” (see Figure 9), it does not connect this preconfigured information in its software with the question posed by the user. All this suggests the need for improvements in the natural language models it can understand and relate to the content of the outlet from which it draws its responses.

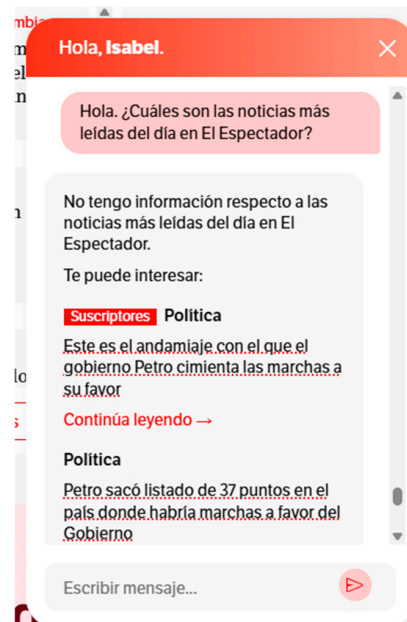


Figure 8. The chatbot responds that it has no information about El Espectador’s most-read news stories, despite having access to that information and offering it when “Check most-read content” is selected.

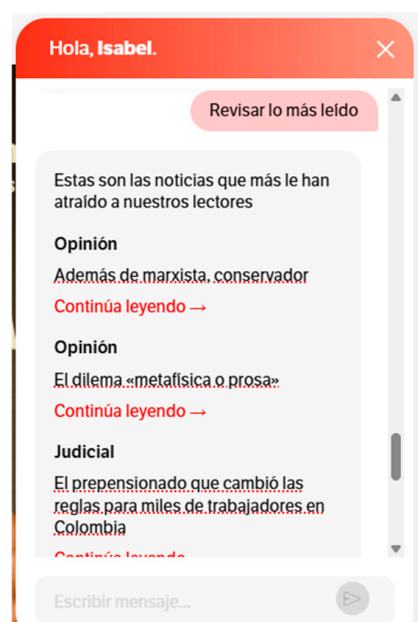


Figure 9. It claims to lack information about El Espectador’s most-read news, even though it can in fact access that data and does provide it when the option “Check most-read content” is selected.

In cases where it cannot provide a valid response, it does display headlines that are presumably of interest to the user and expected to be related to their question; however, these alternative links are random in content: to the question “Tell me the most recent information on Donald Trump”, it replies “I have no information on this topic” and offers as

options a technology news item (“Thousands of users abandon TikTok following censorship complaints against ICE”), which we might consider partially related to Trump, and another from the Bogotá local section that is entirely unrelated: “Mega job fair in Fontibón to offer 5890 vacancies and 1200 companies” (see Figure 10). These are once again dysfunctional behaviours relative to user expectations based on the value proposition announced by El Espectador.

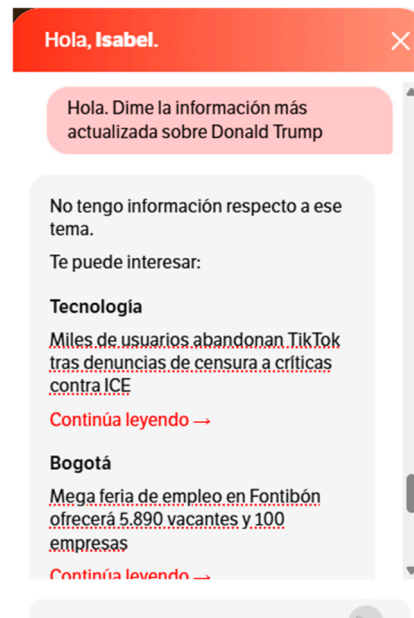


Figure 10. The alternative news items offered by the chatbot are unrelated to the question posed.

As explained above, in order to learn more about chatbots, semi-structured interviews were conducted with the managers of both products. These interviews were analysed according to the categories indicated in Table 2 and yielded the following results, which are also expressed on the basis of these categories (results are summarized in Table 6).

Table 6. Summary of contents by category of analysis.

Category	El País	El Espectador
Media motivation	Added value for subscribers. Technological advancement. Image enhancement	Integration of AI elements. Provide a different, fast experience. Control distribution.
Chatbot objectives	Access to content. Exploring the possibilities of technology.	New distribution channels. Get information without leaving WhatsApp.
Technology used	Open AI-ChatGPT. In-house development. Significant cost. Database: all content from the El País website. Runs on the El País website and its mobile application.	Open AI-ChatGPT. Internal development. Significant cost. Database: all content from the El Espectador website. Runs on WhatsApp. Supports voice searches.

Table 6. *Cont.*

Category	El País	El Espectador
Development and implementation	Project by the Technology field at El País. Maintenance by Product, Editorial and Technology. Tone decided by Editorial and Marketing.	Project by IT Department at El Espectador. Maintenance by Editorial. Effort to “evangelise” and convince the organisation.
Reception by users	Premium subscribers. Four-month trial with select group. Good overall feedback. Use as a search engine and conversational chat.	Subscribers and “super subscribers”. Aimed at the over-50s Approximately 3000 users. “Today’s news”, “recent news”.
Impact on the medium	Improved image. Limited traffic because it is only available to subscribers.	They are analysing which community has the tool. They have better control over distribution.

5.1. Motivation for Implementing the Chatbot

The first category of analysis we used examines the reasons for both media outlets to implement this technology. In the case of El País, we found that the core motivation is related to the image of the media outlet: “To be able to say that we know that technology is here to stay, that we are working with it, that there is a department in charge of this and that we are going to launch products” (Interview 3).

In addition, it is also about offering added value to users who pay a subscription and “rewarding our entire subscriber base who read us, so they can see that, in one way or another, we are trying to do better, not only in terms of content, but also in terms of the product” (Interview 3).

Therefore, this is by no means a definitive, wide-ranging proposal, but rather a tentative move. El País seeks to explore new territory and analyse the possibilities of this technology, while offering its most select audience (premium subscribers) the possibility of having an extra linked to the price they pay for access to the digital newspaper, its content and services.

In the case of El Espectador in Colombia, the development of the chatbot responds to the need to differentiate itself from other newspapers in order to attract and retain a greater number of subscribers, because the intelligent chatbot is perceived as reinforcing the perceived value of the subscription. Beyond increasing revenue for the newspaper, this tool seeks to provide a new information experience for the media’s premium audience: “Readers have less and less time to read, which poses a problem for the newspaper”; therefore, “we wanted to take advantage of the tool to improve the way subscribers can access our content” (Interview 2).

The chatbot helps guide readers in retrieving, consuming and understanding the news, as “there are increasingly complex topics, but less time to devote to them” (Interview 2).

As one of Colombia’s leading news outlets, the approach to creating this assistant also included a certain desire to position itself at the forefront of technology. Thus, without prejudice to its specific objectives, the media outlet sought to “begin integrating elements of artificial intelligence into the website” (Interview 1). The chatbot is one such element, aimed at readers, but at that time El Espectador was already testing other AI functionalities to streamline the work of its journalists, as well as “creating new channels for distributing information” (Interview 1).

As in the case of El País, at no point did they consider a tool with a general scope of consultation, partly because they cannot take risks regarding the accuracy of the information they provide to their readers, given the hallucinations that often plague generative AIs:

“The priority is that all the content suggested by our artificial intelligence comes from El Espectador, because that is the only thing we can control and we can be sure that it is verified content” (Interview 1).

After implementing the chatbot on the newspaper’s website and collecting data on its use by subscribers in the first few months, they considered that the frequency of access data could be improved: “Although it is easy for a digital person to access Elespectador.com from their mobile phone, they do not do so frequently, even if it only takes 5 s, so we decided to use the WhatsApp platform to inform readers” (Interview 2). This consumption accelerator was also accompanied by a more integrated experience in the daily life of the public: “We seek to provide a different, fast experience, designed for people who want to stay informed without leaving WhatsApp, the third most used app in Colombia” (Interview 2).

5.2. Chatbot Objectives

The Objectives category allows us to analyse what each of these media outlets is pursuing with the respective chatbot proposals they have launched. In the case of El País, the objectives are mainly technological. Vera was proposed by the company’s Technology Department, not as a demand from the editorial or marketing fields, although these departments, together with others such as Business Development, later participated in the development and execution of the project.

In this sense, the chatbot’s objective is largely that of a technological demonstrator. The aim is to highlight the possibilities of the technology and observe how the public interacts with it. At the same time, the aim has been to offer a tool that provides added value in terms of content consumption, as it “makes it much easier to access certain information, because, of course, it greatly changes the way a user searches. Before, we were used to searching for ‘war in Ukraine’ and then, once there, you would browse and look for the information you wanted. Now you can ask a specific question and get an exact answer to that question” (Interview 3). For those responsible for the product, the new way of accessing contents “adds considerable value for the user, without having to search piece by piece for what they want” (Interview 4).

For its part, El Espectador’s main objective, almost like a statement of intent, is to “make the content much easier and more digestible for the public, almost like a search tool within the portal” (Interview 1).

They also try to optimise the user experience in their interaction with an AI tool such as the chatbot, creating a “clear interface for people, so that they understand how the chatbot works” (Interview 2). Thus, on the WhatsApp platform, since March 2025, the objective has been to “provide a different, fast experience, designed for people who want to get information without leaving WhatsApp” (Interview 2). Based on this, the application developers worked on “creating an information assistant in which users could ask about recent news, resolve any informational queries, consult the content of El Espectador and obtain a response in natural language plus some suggested links if they wanted further information” (Interview 2).

In any case, the main objective for this second channel of action for the AI chatbot is “to be a tool for obtaining information within WhatsApp, which for us is already an extra channel, and, since our subscribers are using it, we have achieved our goal” (Interview 2).

5.3. Technology Used

The third category of analysis focused on understanding the technology used in both implementation projects. Based on the responses of the managers interviewed, it is possible to understand this process and its technical complexity.

At El País, Vera was developed using the ChatGPT engine, OpenAI's main tool. This Large Language Model was trained using the database of contents published by the newspaper in its nearly 50-year history. In particular, it was trained using contents from the digital edition, which dates back to 1996.

Once the chatbot was trained, its front office was designed, the user interface, which is reminiscent of that of the vast majority of chatbots, with a written dialogue box, since Vera does not recognise voice. In addition, the mode of interaction was defined, which required "modifying certain responses to suit what is wanted" (Interview 3). And finally, the tone with which Vera speaks to the user "is decided between the editorial and marketing teams" (Interview 3).

El Espectador also started from the same technological basis used by the Spanish newspaper: "We use the OpenAI 4 or mini model as a basis, which is quite functional in terms of cost-benefit and capable of providing serious and configurable responses." This is a native development, supported "by the APIs and tools offered by OpenAI for these integrations, such as language models" (Interview 2).

To ensure that the assistant's responses were only configured based on information published by the newspaper, they applied "RAG (Retrieval-Augmented Generation) techniques" in the implementation, thanks to which the language model uses its intelligence, but not its knowledge base, ensuring that the knowledge used to respond is the content of El Espectador (Interview 2).

They also took care to limit its generative pattern, configuring certain parameters "so that it would not be too creative and thus avoid hallucinations." The intention in this regard is for the chatbot to be "aligned with the newspaper's information, customising the language model so that it has El Espectador's own stamp" (Interview 2).

However, they are aware that there is always a certain probability—they estimate it at 1%—that it will offer incorrect results: "That's why we have a team of two people monitoring the responses to make future improvements" (Interview 2).

5.4. Implementation Process

This category of analysis has provided a better understanding of the development and implementation process of the two chatbots analysed. In the case of El País, we found that the impetus to launch the tool came from the Technology department, which wanted to explore the possibilities of the agreement signed with OpenAI: "It has been an internal project involving many fields, including Technology, Product Design, Editorial, and Business. In other words, practically all fields within the group have been involved" (Interview 4).

The first version of Vera was put into production and underwent a two-month testing period. During that time, a select group of users acted as beta testers and put the tool's capabilities to the test. During this period, the development team surveyed users twice in order to gather as much information as possible to refine the product. Once the necessary adjustments had been made, the Marketing and Editorial teams worked on defining Vera's look and feel: what the interaction interface should be like, how it would appear and where it would be located, and how the user would interact with the tool. The final step was to adjust the tone, the 'voice' with which Vera addresses the person making enquiries. In this case, El País decided to give it a warm but serious tone, not too informal.

Once the chatbot was up and running, El País did not substantially improve the proposal, as the technology development team prioritises other products. They simply monitor Vera's responses to ensure they are correct and, if they are not, try to improve the user experience. However, for the time being, the tool is not being developed further.

At the same time, it is important to mention the cost, which is high: “Each question asked has an associated cost, so commissioning involves not only development costs, nor just the maintenance and storage costs required, in addition to the database, but also each question asked by the user has a cost for the group” (Interview 3). This cost is money that El País pays to OpenAI for each query made by users.

In the case of El Espectador, the initial proposal came from the newspaper’s New Technologies Department, whose director sent a proposal to the Steering Committee, the Editor-in-Chief of El Espectador, Fidel Cano, and the managing director, Angélica de Lagos, to propose this new experience. His initiative sparked a fruitful debate among the newspaper’s senior management: “The initial debates revolved around the risks present in language models, such as hallucinations... Part of the work was to educate managers and editors to make them understand that these tools are highly configurable” (Interview 2).

Once the project was passed, El Espectador’s multimedia editor took charge of shaping the chatbot so that it would work in terms of user experience and understanding of the interface by the service’s target audience. “Most of our subscribers are over 50 years old, so the tool had to deliver content to them in an even easier and much more digestible way” (Interview 1).

They followed an internal start-up methodology, first developing an MVP (minimum viable product): “We managed to implement several prototype products that allowed us to provide a preliminary experience to find out if the product was viable before rolling it out into production and scaling it up” (Interview 2). Once that first level of development was complete, they decided to implement the tool in its entirety and consulted with external suppliers, although they ultimately decided to carry out the development internally with El Espectador’s New Technologies team: “It took us about a year to develop it, mainly because a large part of the process involved research and analysing the proposal in great detail. We knew that launching it could be very sensitive, because it was a new and unique product in Colombia and because we couldn’t risk El Espectador’s reputation” (Interview 2).

Before putting the tool into production, the team made sure that it met the characteristics stipulated at the outset and, after testing, “we opened the chatbot, initially to so-called ‘super subscribers’, because we didn’t know if it would be widely adopted. After a while, we opened it up to all subscribers” (Interview 1).

The chat offers certain closed options, one of which is generative (“Ask me and I’ll give you a quick summary”) designed for the public to ask open-ended questions—“What’s happening in Colombia today?”, “What about security?”-. When the subscriber asks the question, the chatbot returns a summary based on the information published on the El Espectador website. It also provides links to articles published in the newspaper so that readers can go deeper into the information. To optimise the process with reader feedback, readers can rate the responses with a star rating system.

Its developers still consider it a “rather cold product; it is very direct, we haven’t given it a personality”; in fact, it doesn’t even have a name, unlike other assistants. That is why they consider it “a basic product that will allow us to later create products more focused on certain niches” (Interview 2).

As for the impact on costs that this type of technology has had on the media, they consider that they have managed it efficiently by developing it internally rather than resorting to an external company, with the added benefit of greater control over the present and future of the assistant. From this perspective, through their market research, they estimate that “the cost of the tool has fallen twice since we started implementing it, and we believe it will continue to fall because it is a global trend” (Interview 2).

5.5. Public Reception

The fifth category of analysis applied to the materials collected allows us to understand how these chatbots have been received by the public.

Vera, the product launched by El País, is only available to premium users, which is a limited group. However, its use allows us to assess how they use the tool: “You find all kinds of cases: from people who try to use it as if it were ChatGPT, which in the end it is not, because it is not a standard assistant, to people who use it as a search engine, as it also offers a list of articles written by I don’t know who.” And also questions about what is happening in the war in such and such a place, who was the President of such and such, or when was the last time. In other words, in general, they ask questions at that level” (Interview 3).

In other words, users have two core types of queries: on the one hand, they use Vera to replace the typical search function, letting the artificial intelligence assistant find the content and offer it to them in groups. On the other hand, they ask questions that the chatbot answers in its own reworkings based on the media content.

At this point, our observation has identified the difficulties that the chatbot encounters in consistently providing elaborated responses to apparently straightforward questions, and how these responses can vary depending on the day or time of day (see Figure 11). This divergence between what the project managers indicate and the data from our observation is notable, yet equally difficult to understand. It appears to be related to the internal logic of the language model and does not correspond to a technical decision on the part of the human managers. For example, it is difficult to understand why Vera does not provide football results in the morning but does so in the afternoon, when the query is exactly the same and only the time of day differs. Gaining detailed knowledge of why this situation occurs would require access to the core of OpenAI’s technology, something that this research has not achieved.

Vera

¡Hola! Gracias por tu mensaje. Actualmente, no tengo información específica sobre los últimos resultados de la Liga de Fútbol. Sin embargo, puedo ayudarte a buscar información sobre otros temas relacionados con el fútbol o cualquier otra consulta que tengas. ¿Te gustaría saber algo más?

- La hora de Álex Baena:** Desde que reapareció en enero el fichaje estrella del Atlético este verano no pasa de 60 minutos cuando es titular y sin Barrios, lesionado, es clave que recupere el nivel ofrecido antes de la lesión. [Leer más >](#)
- Lindsey Vonn es operada por tercera vez: “Sé que me recuperaré”:** La estadounidense, que participó en los Juegos de Milán-Cortina con el cruzado roto, sufrió una caída en la final de descenso de esquí alpino y tuvo que ser evacuada en helicóptero. [Leer más >](#)
- Daniel Milagros estrena el historial de España en el patinaje de velocidad de los Juegos Olímpicos de invierno:** El navarro, primer español en la historia olímpica de la disciplina, termina en 27ª posición la final de los

Vera

¡Hola! A continuación, estos son los resultados disponibles de la Liga de Fútbol:

- Elche - Osasuna: viernes 21.00, Dazn
- Espanyol - Celta: sábado 14.00, Movistar+
- Getafe - Villarreal: sábado 16.15, Dazn
- Sevilla - Alavés: sábado 18.30, Teledeporte
- Real Madrid - Real Sociedad: sábado 21.00, Movistar+
- Mallorca - Betis: domingo 21.00, Dazn
- Rayo - Atlético: domingo 16.15, Dazn
- Levante - Valencia: domingo 18.30, Movistar+
- Girona - Barcelona: lunes 21.00, Dazn

Son muchos, puedes encontrar más ejemplos en los siguientes artículos. ¿Quieres ver más resultados sobre el tema o puedo ayudarte con algo más?

- Nikola Topic debuta en la NBA cuatro meses después de haber sido diagnosticado con cáncer de testículos:** El base serbio de 20 años se estrena con los Oklahoma City Thunder tras haber superado una operación y tres rondas de quimioterapia entre octubre y diciembre. [Leer más >](#)

Figure 11. Example of the same query made in the morning session (left) and afternoon session (right) on the same day.

Nevertheless, and despite these possible shortcomings, the managers at El País consider that the product has been very well received, although the volume of usage is still limited and does not yet allow for clear conclusions to be drawn in this regard.

At El Espectador in Colombia, initial data shows that “readers use it a lot to keep up to date: for example, ‘today’s news’ and ‘recent news’ are the most frequent queries of all, and they are made about three or four times a day” (Interview 2).

In terms of the format of queries, “voice notes are proving useful, especially for quick questions, such as ‘tell me if there are any traffic restrictions today’ or ‘why is there a traffic jam on such-and-such a day in Bogotá” (Interview 1).

From the comments and assessments shared by readers, it is clear that they would like to receive more extensive responses: “Summaries of a thousand characters may be a little sparse for them, as they are used to articles of a thousand words, which is why they make successive queries; for example, in the news, after the first query, others appear such as ‘tell me more about this news item’ or ‘expand on this detail” (Interview 2).

The analysis, although still incomplete, continues to be carried out based on current data, because “we first launched the chatbot on the website and then on the WhatsApp channel, and, moreover, initially it was only for a small group of subscribers, it was not open to everyone” (Interview 1).

5.6. *Impact on the Media*

The last category of analysis applied to the research materials provides information for analysing the extent to which chatbots impact the media and their performance.

In the case of El País, the impact is still very limited. As it is a small group of users, it seems difficult to assess what this means in terms of web metrics: increased traffic, user retention, etc.

Therefore, at the time of completing this research, there is no evidence of a significant impact beyond a possible improvement in El País’s image, which is also difficult to measure.

Finally, at El Espectador, those responsible continue to measure and analyse data to understand “how much of the audience on the website is replicated on WhatsApp and how the two channels work” (Interview 1). The interpretation of this data will be core for making future decisions at El Espectador: “We have user data, which allows us to segment so that we can send them targeted advertising, offer personalised content or redirect them to other traffic sources (including links to the website, YouTube links, etc.) so that all channels complement each other” (Interview 2). Changes are also expected in subscription levels. Users who exhaust their 20 daily queries “would be willing to pay for a more comprehensive tool, so we could have a standard assistant for basic subscribers and perhaps a second paid assistant with unlimited queries for those who use the service more intensively” (Interview 2).

They believe that the final impact is undoubtedly positive for El Espectador, which with this AI-based chatbot has managed to “build its own highly efficient and automatable channel” (Interview 2). Given the decline in referral traffic that until recently was provided by search engines and social networks, “media outlets around the world are currently very interested in creating their own channels where we have control over the distribution of our content, and these types of tools are a way to achieve this” (Interview 2).

6. Discussion

The results analysed in the previous section allow us to consider that the three main objectives set at the beginning of this research have been achieved. Furthermore, they offer a satisfactory answer to the research question, which, as stated above, sought to explore the emerging field of conversational chatbots in the media.

In this regard, we find that, so far, the use of these tools in media outlets such as *El País* and *El Espectador* has two core motivations: to improve the user experience and to experiment with the integration of artificial intelligence elements. This experimentation provides the media with knowledge for future actions and also an image of being at the forefront of technology, which reinforces their position in the market. This motivation is consistent with that found by other authors who have analysed innovation in digital media, such as [López Hidalgo and Ufarte Ruiz \(2019\)](#), who, in studying the Labs of some Spanish media outlets, find that they serve “to experiment in the creation of products and services in order to meet the challenges of the future”. [Imre and Wenger \(2020\)](#) have also agreed that the influence of companies is a key factor in the adoption of technological innovations in the media. Closely related to this idea is a motivation linked to marketing: offering subscribers something different and, therefore, giving them reasons to continue their subscriptions. This is related to what [Molinillo and Japutra \(2017\)](#) pointed out when studying the reasons for the adoption of technologies in organisations: “ICT adoption allows businesses to be more competitive and increase their performance, particularly when they use ICT for marketing-related functions”. [Wu \(2026\)](#) also identifies interesting uses of chatbots in newsrooms: “AI tools have enabled journalists to relocate related stories that they can use to cross-promote their pieces”.

Beyond motivations, research also allows us to understand how audiences and the media themselves are using this technology. In this sense, the introduction of artificial intelligence in the implementation of chatbots has transformed these conversational assistants. The first generation of chatbots, built on conventional programming, showed that the public was surprised by these tools and even developed affectionate relationships with them ([Gómez-Zará & Diakopoulos, 2020](#)). This was because their “light-hearted way of reporting the news and interacting with the audience surprises users, who get caught up in the game of answering the bot, following its news itineraries and sharing its content” ([Herrero-Diz & Varona-Aramburu, 2018](#)). In the first chatbots with artificial intelligence, [Ischen et al. \(2020\)](#) already found something similar when they pointed out “the crucial role of user’s enjoyment in the persuasion context”.

In contrast, current users have already moved beyond that initial phase of surprise and seem to be making much more systematic use of chatbots, focusing on searching for content and analysing information to obtain specific answers. This is where the ability to converse with the tool comes into play, multiplying the possibilities for interaction and, therefore, for exploring content. That is, capacity to retain the user, in line with what [Gao et al. \(2025\)](#) state when explaining that their study “emphasises the paramount role of cultivating consumer trust in chatbot capabilities to ensure customer retention”. Nevertheless, we also find that although the chatbots are quite effective at responding, they still make errors and there are moments when they fail to answer the questions posed by users, even when these are always the same.

Aside from these shortcomings, that advanced and specialised knowledge displayed by users is surely the result of their experience with the numerous chatbots already available on the Internet. These tools have enabled the public to become much more adept at handling artificial intelligence and to use it in a more natural and routine manner, even when engaging with it within a conventional media outlet.

In this regard, the work of [Herrero-Diz and Varona-Aramburu \(2018\)](#) correctly predicted the progress from conventional programming to the possibilities of artificial intelligence. That work announced that, even then, the media were working “on processes to develop natural language to simulate human conversations and encourage interaction with the user”. A similar stage of development was captured by [Jones and Jones \(2019\)](#) when researching the chatbots that had been developed by the BBC at that time. These

authors already pointed out that these tools “are changing the way journalists, managers and developers think about what is possible and may be altering the expectations that audiences have of their engagement with the BBC and news more generally.”

In our research, however, we found that those expectations are now a thing of the past and that the public and media professionals naturally accept the new model of relationship between audience and media. The big change is that, in those early chatbots, media professionals had to decide and programme the contents that the robot would release (Ford & Hutchinson, 2019; Lokot & Diakopoulos, 2016), whereas now artificial intelligence takes care of that task, relieving journalists of the burden and allowing them to simply monitor that the responses are appropriate and the tone of the chatbot is correct. In this way, artificial intelligence automates yet another task in newsrooms, continuing a process that authors such as Linden (2017) and Grimme (2021) have been describing for years.

This automation scenario will reinforce another of the findings detected by this research: artificial intelligence chatbots offer digital media an interesting way to make much better use of their content archives. This exploitation will be automatic, based on user queries and the decisions made by the chatbot. Thus, our work aligns with the findings of Bilińska (2020), who pointed out that chatbots give users “quick access to content on the topic they are interested in, without having to open a website in the search engine or a dedicated mobile application.” In other words, it changes access to the content repository, which will be based on the conversation between the user and the chatbot, reducing traditional searches and competing with the possible selection of present and past content made by the field of editorial. This new way of interacting must therefore be understood as competition for the two conventional channels: search and journalistic supply. As authors such as Sánchez Gonzales and Sánchez González (2017) have already pointed out, chatbots may even “replace search engines as the natural gateway to the Internet” and, therefore, to the media itself.

However, it should be noted that this moment of competition still seems a long way off. For now, in terms of volume of use, we find, as in practically all studies of first-generation chatbots, that public use is still low. This is mainly because the two media studied offer their developments only to a small group of premium users.

It remains to be seen what will happen when they are offered to the general public, something that, at present, still seems limited by cost. Judging by the enthusiastic adoption of generalist chatbots (ChatGPT, Gemini, Claude, etc.), it is reasonable to predict that their use in the media will also merit. However, there is still no certainty in this regard, beyond the positive reception that both developments have received among users, according to the results of our research.

This exploratory work also shows that the arrival of artificial intelligence chatbots does not seem to arouse much interest among journalists in the media studied. Beyond the fact that some teams are involved in defining and monitoring these assistants, there is no discernible impact among editors. This opens up a line of future research: to find out how professionals relate to the chatbots implemented by their own media outlets.

In addition, it will also be useful to explore how the relationship between audiences, chatbots and media evolves in the future. The way in which users employ the chatbot may change as the tool becomes more sophisticated and commits fewer errors, thereby proving much more useful. If the results offered by chatbots continue to improve, it is possible that content consumption will decline, replaced by texts generated by artificial intelligence. This is expected to lead to changes in the media at many levels, but especially in commercial and organisational terms. Analysing these transformations will be a task for future research.

Author Contributions: Conceptualisation, G.S.-M., I.G.C. and D.V.A.; methodology, G.S.-M. and D.V.A.; investigation, I.G.C. and D.V.A.; resources, G.S.-M.; writing—original draft preparation, G.S.-M., I.G.C. and D.V.A.; writing—review and editing, I.G.C.; supervision, G.S.-M. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. Given the minimal risk posed to participants and the absence of invasive procedures or the collection of sensitive information, prior review by the University's Ethics Committee was not required. All participants were fully informed about the purpose of the study, the confidentiality of their data, and their right to withdraw at any time. Informed consent was obtained from all participants prior to their involvement. No personal or identifiable data have been disclosed or published at any stage of the research process or in this manuscript.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors on request.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Guidelines for the semi-structured interviews: thematic blocks

Intro: Please tell us about your professional background and your current position at El Espectador/El País.

BLOCK 1—Description of the chatbot: understanding its structure, functionality, and orientation

Examples of questions for Block 1:

What was the process behind the creation of this chatbot? Why did you decide to launch it? When was it created, who designed it, programmed it, and who is responsible for its maintenance?

Bot identity: name, tone, and voice used to address readers; graphic elements that define its visual appearance.

Chatbot behavior and operation.

Do your competitors have something similar? Does having it provide El Espectador/El País with a competitive advantage?

BLOCK 2—Technology

Understanding the technology used in the chatbot's design and development.

Examples of questions for Block 2:

What technology and type of AI underpin it?

Creation and maintenance costs.

How was its launch communicated—through the outlet itself, on social media, via advertising campaigns, or through email announcements?

BLOCK 3—Audience impact

This section aims to understand how the chatbot influences the relationship between the news outlet and its audiences.

Examples of questions for Block 3:

What does it contribute to the outlet in terms of brand image, time savings, or revenue generation?

How do users engage with it?

What impact does it have on the news outlet's website traffic?

What feedback do readers provide regarding its performance?

References

- Adams, J. (2024). Examining ethical and social implications of digital mental health technologies through expert interviews and sociotechnical systems theory. *Digital Society*, 3, 24. [CrossRef]
- Ali, M., Wood-Harper, T., & Wood, B. (2024). Understanding the technical and social paradoxes of learning management systems usage in higher education: A sociotechnical perspective. *Systems Research and Behavioral Science*, 41(1), 134–152. [CrossRef]
- Alonso, L. E. (1999). Subject and discourse: The place of the open interview in qualitative sociology practices. In J. Delgado, & J. Gutiérrez (Eds.), *Qualitative research methods and techniques in social sciences* (pp. 225–239). Editorial Síntesis.
- Apablaza-Campos, A., Bonilla, L. C., & Lopezosa, C. (2025). Contents SEO and artificial intelligence: Experiences in digital media. In *The integration of artificial intelligence in digital communication* (pp. 335–349). Tirant Humanidades.
- Belair-Gagnon, V., Lewis, S., & Agur, C. (2020). Failure to launch: Competing institutional logics, intrapreneurship, and the case of chatbots. *Journal of Computer-Mediated Communication*, 25(4), 291–306. [CrossRef]
- Bilińska, P. (2020). Chatbots as a content distribution tool used by media publishers. *Studia Medioznawcze*, 21–23(82), 689–707. [CrossRef]
- Cerón-Guzmán, J. A., & León-Guzmán, E. (2016). A sentiment analysis system of Spanish tweets and its application in Colombia 2014 presidential election. In *2016 IEEE international conferences on big data and cloud computing (BDCloud), social computing and networking (SocialCom), sustainable computing and communications (Sustain-Com) (BDCloud-SocialCom-SustainCom)* (pp. 250–257). IEEE. [CrossRef]
- Chand, S. P. (2025). Methods of data collection in qualitative research: Interviews, focus groups, observations, and document analysis. *Advances in Educational Research and Evaluation*, 6(1), 303–317. [CrossRef]
- Chan-Olmsted, S. M. (2019). A review of artificial intelligence adoptions in the media industry. *International Journal on Media Management*, 21(3–4), 193–215. [CrossRef]
- Chuan, C. H., Tsai, W. H. S., & Cho, S. Y. (2019). Framing artificial intelligence in American newspapers. In *the 2019 AAAI/ACM conference on AI, ethics, and society* (pp. 339–344). Association for Computing Machinery. [CrossRef]
- Cools, H., & Diakopoulos, N. (2024). Uses of generative AI in the newsroom: Mapping journalists' perceptions of perils and possibilities. *Journalism Practice*, 20(3), 878–896. [CrossRef]
- Dahlman, S., Gulbrandsen, I. T., & Just, S. N. (2021). Algorithms as organizational figuration: The sociotechnical arrangements of a fintech start-up. *Big Data & Society*, 8(1). [CrossRef]
- De-Lima-Santos, M. F., & Ceron, W. (2022). Artificial intelligence in news media: Current perceptions and future outlook. *Journalism and Media*, 3, 13–26. [CrossRef]
- Dengel, A., Gehrlein, R., Fernes, D., Görlich, S., Maurer, J., Pham, H. H., Großmann, G., & Eisermann, N. D. (2023). Qualitative research methods for large language models: Conducting semi-structured interviews with ChatGPT and BARD on computer science education. *Informatics*, 10, 78. [CrossRef]
- Denny, E., & Weckesser, A. (2022). How to do qualitative research? Qualitative research methods. *BJOG: An International Journal of Obstetrics and Gynaecology*, 129(7), 1166–1171. [CrossRef]
- Díaz Noci, J. (2023). Artificial intelligence, news and the media: A legal approach from the perspective of intellectual property to the concept and attribution of authorship. *Textual & Visual Media*, 17(1), 7–21. [CrossRef]
- Elias, N. (1995). Technization and civilization. *Theory, Culture & Society*, 12(3), 7–42. [CrossRef]
- El País. (2024). *OpenAI announces agreement with Prisa Media and Le Monde*. El País. Available online: <https://elpais.com/comunicacion/2024-03-13/open-ia-anuncia-un-acuerdo-con-prisa-media-y-le-monde.html> (accessed on 5 April 2025).
- El País. (2025). *El País launches vera El País, a conversational assistant with artificial intelligence*. El País. Available online: <https://elpais.com/comunicacion/el-pais-que-hacemos/2025-03-26/el-pais-lanza-vera-el-pais-un-asistente-conversacional-con-inteligencia-artificial.html> (accessed on 5 April 2025).
- Fernández Barrero, Á., & López Redondo, I. (2022). Verification in the era of fake news. Some examples related to COVID-19. *Ámbitos: International Journal of Communication*, 57, 124–137. [CrossRef]
- Ford, H., & Hutchinson, J. (2019). Newsbots that mediate journalist and audience relationships. *Digital Journalism*, 7(8), 1013–1031. [CrossRef]
- Gao, J., Opute, A. P., Jawad, C., & Zhan, M. (2025). The influence of artificial intelligence chatbot problem solving on customers' continued usage intention in e-commerce platforms: An expectation-confirmation model approach. *Journal of Business Research*, 200, 115661. [CrossRef]
- García-Avilés, J. A. (2021). Review article: Journalism innovation research, a diverse and flourishing field (2000–2020). *Information Professional*, 30(1), e300110. [CrossRef]
- Garriga, M., Ruiz-Incertis, R., & Magallón-Rosa, R. (2024). Artificial intelligence, disinformation and proposals for literacy around deepfakes. *Observatorio (OBS*)*, 18(5), 175–194. [CrossRef]
- Google. (2025, October 7). *Google search: AI mode comes to Spain*. Google Blog. Available online: <https://blog.google/intl/es-es/productos/la-busqueda-de-google-el-modo-ia-llega-a-espana/> (accessed on 9 March 2026).

- Gómez-Zar4, D., & Diakopoulos, N. (2020). Characterising communication patterns between audiences and newsbots. *Digital Journalism*, 8(9), 1093–1113. [CrossRef]
- Grimme, M. V. (2021). Factors influencing the rejection of automated journalism: A systematic literature review. *Nordic Journal of Media Management*, 2(1), 3–21. [CrossRef]
- Hepp, A., Loosen, W., Dreyer, S., Jarke, J., Kannengießer, S., Katzenbach, C., Malaka, R., Pfadenhauer, M., Puschmann, C., & Schulz, W. (2023). ChatGPT, LaMDA, and the hype around communicative AI: The automation of communication as a field of research in media and communication studies. *Human-Machine Communication*, 6, 41–63. [CrossRef]
- Herrero-Diz, P., & Varona-Aramburu, D. (2018). Use of chatbots to automate information in the Spanish media. *El Profesional de la Información*, 27(4), 742–749. [CrossRef]
- Howard, M. (2025, November 10). *The story behind the TIME AI agent*. Time. Available online: <https://time.com/7332572/the-story-behind-the-time-ai-agent/> (accessed on 15 November 2025).
- Husband, G. (2020). Ethical data collection and recognising the impact of semi-structured interviews on research respondents. *Education Sciences*, 10(8), 206. [CrossRef]
- Imre, I., & Wenger, D. (2020). Where newsroom leaders see technology facilitating innovation in local TV news. *Electronic News*, 14(4), 151–167. [CrossRef]
- Ioscote, F., Gonçalves, A., & Quadros, C. (2024). Artificial intelligence in journalism: A ten-year retrospective of scientific articles (2014–2023). *Journalism and Media*, 5(3), 873–891. [CrossRef]
- Ischen, C., Araujo, T., van Noort, G., Voorveld, H., & Smit, E. (2020). “I am here to assist you today”: The role of entity, interactivity and experiential perceptions in chatbot persuasion. *Journal of Broadcasting and Electronic Media*, 64(4), 615–639. [CrossRef]
- Jones, B., & Jones, R. (2019). Public service chatbots: Automating conversation with BBC news. *Digital Journalism*, 7(8), 1032–1053. [CrossRef]
- Junnier, F. (2024). Action and understanding in the semi-structured research interview: Using CA to analyse European research scientists’ attitudes to linguistic (dis)advantage. *Journal of English for Academic Purposes*, 66, 101355. [CrossRef]
- Kallio, H., Pietilä, A., Johnson, M., & Kangasniemi, M. (2016). Systematic methodological review: Developing a framework for a qualitative semi-structured interview guide. *Journal of Advanced Nursing*, 72(12), 2954–2965. [CrossRef]
- Knott, E., Rao, A., Summers, K., & Teeger, C. (2022). Interviews in the social sciences. *Nature Reviews Methods Primers*, 2, 73. [CrossRef]
- Lewis, S. C., Guzman, A. L., & Schmidt, T. R. (2019). Automation, journalism, and human–machine communication: Rethinking roles and relationships of humans and machines in news. *Digital Journalism*, 7(4), 409–427. [CrossRef]
- Lin, C.-C., Huang, A. Y. Q., & Yang, S. J. H. (2023). A review of AI-driven conversational chatbots implementation methodologies and challenges (1999–2022). *Sustainability*, 15, 4012. [CrossRef]
- Linden, C. G. (2017). Decades of automation in the newsroom: Why are there still so many jobs in journalism? *Digital Journalism*, 5(2), 123–140. [CrossRef]
- Lokot, T., & Diakopoulos, N. (2016). News bots: Automating news and information dissemination on Twitter. *Digital Journalism*, 4(6), 682–699. [CrossRef]
- Londoño-Proañ0, C., & Buele, J. (2025). Can artificial intelligence replace journalists? A theoretical approach. *Frontiers in Communication*, 10, 1537146. [CrossRef]
- López Hidalgo, A., & Ufarte Ruiz, M. J. (2019). Laboratorios de periodismo en España. Nuevas narrativas y retos de futuro. *Ámbitos. Revista Internacional De Comunicación*, 1–11. [CrossRef]
- Manisha, R., & Acharya, K. (2023). The impact of artificial intelligence on news curation and distribution: A review literature. *Journal of Communication and Management*, 2(01), 23–26. [CrossRef]
- Mayoral-Sánchez, J., Mera-Fernández, M., & Morata-Santos, M. (2024). Chapter 8. Integration of artificial intelligence in newsrooms: The experience of the media in Spain. *Espejo De Monografías De Comunicación Social*, 25, 187–209. [CrossRef]
- Mayoral-Sánchez, J., Parratt-Fernández, S., & Mera-Fernández, M. (2023). Journalistic use of AI in Spanish media: Current map and prospects for the immediate future. *Studies on the Journalistic Message*, 29(4), 821–832. [CrossRef]
- Mayring, P. (2014). *Qualitative content analysis. Theoretical foundation, basic procedures and software solution*. Autoedition.
- McIntosh, M. J., & Morse, J. M. (2015). Situating and constructing diversity in semi-structured interviews. *Global Qualitative Nursing Research*, 2. [CrossRef] [PubMed]
- McKelvey, F., & Dubois, E. (2017). *Computational propaganda in Canada: The use of political bots*. (Computational Propaganda Project. Working Paper No. 2017.6). University of Oxford. Available online: <http://comprop.oii.ox.ac.uk/wp-content/uploads/sites/89/2017/06/Comprop-Canada.pdf> (accessed on 22 November 2025).
- Miguel, M. C., & Cabrera, B. D. (2020). Perspectives on Chatbot technologies and their application to language assessment interviews. *Pragmalinguistics*, 2, 100–113.
- Molinillo, S., & Japutra, A. (2017). Organisational adoption of digital information and technology: A theoretical review. *The Bottom Line*, 30(1), 33–46. [CrossRef]

- Moran, R., & Jawaid, S. (2022). Robots in the news and newsrooms: Unpacking meta-journalistic discourse on the use of artificial intelligence in journalism. *Digital Journalism*, 10(10), 1756–1774. [CrossRef]
- Niño, M., & Borges, G. (2025). Artificial intelligence and algorithms in journalism: Transformations in professional culture and university training. *RAE-IC, Journal of the Spanish Association for Communication Research*, 12, raeic12e04. [CrossRef]
- Noain-Sánchez, A. (2022). Addressing the impact of artificial intelligence on journalism. *Communication & Society*, 35(3), 105–121. [CrossRef]
- Nordberg, O., & Guribye, F. (2023). Conversations with the News: Co-speculation into Conversational Interactions with News Content. In *ACM conference on conversational user interfaces (CUI '23)*, July 19–21, 2023, Eindhoven, The Netherlands. ACM. [CrossRef]
- Oh, S., & Jung, J. (2025). Harmonising traditional journalistic values with emerging AI technologies: A systematic review of journalists' perception. *Media and Communication*, 13, 9495. [CrossRef]
- Pano, A. (2012). Dialogue and conversational information in the Spanish digital press. In *Il dialogo. Lingue, letteratura, linguaggi, culture. Proceedings of the XXV AISPI conference, Naples, Italy, 18–21 February 2009* (pp. 351–358). AISPI Edizioni.
- Parratt-Fernández, S., Mayoral-Sánchez, J., & Mera-Fernández, M. (2021). Application of artificial intelligence to journalism: Analysis of academic production. *Information Professional*, 30(3), e300317. [CrossRef]
- Perreault, G., & Ohme, J. (2025). ChatBots as artificial intermediaries? Adaptation to artificial intelligence in newsrooms. *Journalism Studies*, 26(15), 1914–1935. [CrossRef]
- Pérez-Montoro, M., & Lopezosa, C. (2026). Generative AI in data visualisation: Usefulness, challenges and comparative assessment. *Revista Mediterránea De Comunicación*, 17(1), E30124. [CrossRef]
- Pérez-Seijo, S., Gutierrez-Caneda, B., & Lopez-Garcia, X. (2020). Digital journalism and high technology: From consolidation to renewed challenges. *Index Comunicación*, 10(3), 129–152. [CrossRef]
- Rentoul, J. (2020). *Will artificial intelligence put journalists out of work?* The Independent. Available online: <https://www.independent.co.uk/independentpremium/editors-letters/artificialintelligence-journalism-reporting-robots-transcribing-news-digital-a9337451.html> (accessed on 11 November 2025).
- Rodríguez-Pérez, C. (2021). Online disinformation and fact-checking in social polarization contexts: The fact-checking journalism of Colombiacheck, La Silla Vacía and AFP during the 21N national strike in Colombia. *Estudios sobre el Mensaje Periodístico*, 27(2), 623–637. [CrossRef]
- Romera-Fadón, J. A., Cristòfol Rodríguez, J., & Peláez Agudo, D. (2024). IA: Motor de cambio en El Español y sus redes sociales. *Miguel Hernández Communication Journal*, 15(2), 193–208. [CrossRef]
- Sarrionandia, B., Peña-Fernández, S., & Pérez-Dasilva, J. A. (2025). Artificial intelligence as a driver of innovation in the media: Technical appropriation, informational caution. *Studies on the Journalistic Message*, 31(3), 603–613. [CrossRef]
- Sánchez Gonzales, H., & Sánchez González, M. (2017). Bots as a news service and emotional connection with audiences. The case of Politibot. *Doxa Comunicación. Interdisciplinary Journal of Communication and Social Science Studies*, 25, 63–84. [CrossRef]
- Tejedor, S., & Vila, P. (2021). Exo journalism: A conceptual approach to a hybrid formula between journalism and artificial intelligence. *Journalism and Media*, 2(4), 830–840. [CrossRef]
- Terol, T. M. (2023). Media innovation: Applications of artificial intelligence in journalism in Spain. *Textual & Visual Media*, 17(1), 41–60. [CrossRef]
- Trapova, A., & Mezei, P. (2022). Robojournalism—A copyright study on the use of artificial intelligence in the European news industry. *GRUR International*, 71(7), 589–602. [CrossRef]
- Trejos-Gil, C. A., & Gómez-Monsalve, W. D. (2024). Artificial intelligence in media and journalism. Systematic review on Spain and Latin America in Scopus and Web of Science databases (2018–2022). *Palabra Clave*, 27(4), e2741. [CrossRef]
- Trillo-Domínguez, M., & Alberich-Pascual, J. (2017). Deconstruction of journalistic genres and new media: From the inverted pyramid to the Rubik's cube. *Information Professional*, 26(6), 1091–1099. [CrossRef]
- Túñez-López, J. M., Fieiras Ceide, C., & Vaz-Álvarez, M. (2021). Impact of Artificial Intelligence on Journalism: Transformations in the company, products, contents and professional profile. *Communication & Society*, 34(1), 177–193. [CrossRef]
- Ufarte-Ruiz, M. J., Murcia-Verdú, F. J., & Túñez-López, J. M. (2023). Use of artificial intelligence in synthetic media: First newsrooms without journalists. *Profesional de la Información*, 32(2), e320203. [CrossRef]
- Vara-Miguel, A. (2025). *Executive report 2025. Digital news report Spain* [Blog Entry]. University of Navarra. Available online: <https://www.unav.edu/web/digital-news-report/busqueda/-/blogs/informe-ejecutivo-2025> (accessed on 5 November 2025).
- Vega, C. (2024). *El Espectador launches artificial intelligence chatbot for Premium subscribers*. El Espectador. Available online: <https://www.elespectador.com/suscriptores/el-espectador-lanza-chatbot-con-inteligencia-artificial-para-suscriptores-premium/> (accessed on 8 April 2025).
- Webster, J. (2025). Sociotechnical imaginaries in a postdigital world: Teachers' perceptions of digital citizenship education. *Postdigital Science and Education*, 7, 770–787. [CrossRef]
- Wheatley, D. (2021). Tracking beyond the text: The use of regressive content analysis to study journalistic practices. *Communication & Methods*, 3(1), 40–55. [CrossRef]

- Wu, S. (2026). Journalists as individual users of artificial intelligence: Examining journalists' "value-motivated use" of ChatGPT and other AI tools within and without the newsroom. *Journalism*, 27(2), 388–406. [[CrossRef](#)]
- Xiao, Q., Hu, Q., Xiao, J., Cao, H., & Shen, H. (2025). Can generative AI move from individual use to collaborative work? Experiences, challenges and opportunities of integrating GenAI into collaborative newsroom routines. *arXiv*, arXiv:2509.10950. [[CrossRef](#)]
- Xie, Z. (2021). Impact of artificial intelligence on new media operations and communication. In J. Macintyre, J. Zhao, & X. Ma (Eds.), *The 2021 international conference on machine learning and big data analytics for IoT security and privacy. SPIoT 2021. Lecture notes on data engineering and communications technologies* (Vol. 97). Springer. [[CrossRef](#)]
- Zanzotto, F. (2019). Human-in-the-loop artificial intelligence. *Journal of Artificial Intelligence Research*, 64(1), 243–252. [[CrossRef](#)]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.